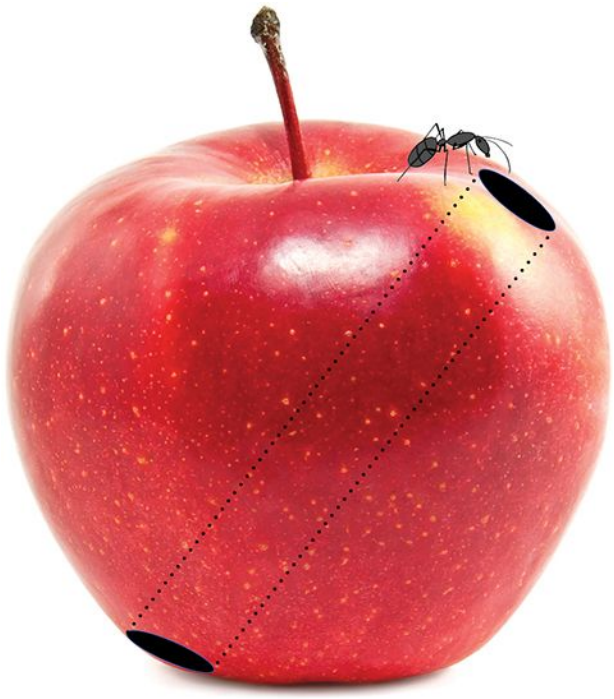


Fig. 14.1. An ant explores a wormhole-endowed apple.

The apple's delicious interior, through which the wormhole passes, is not part of the ant's universe. It is a three-dimensional bulk or hyperspace (Chapter 4). The wormhole's wall can be thought of as part of the ant's universe. It has the same dimensionality as the universe (two dimensions) and it joins onto the universe (the apple's surface) at the wormhole's entrance. From another viewpoint, the wormhole's wall is *not* part of the ant's universe; it is just a shortcut by which the ant can travel across the bulk, from one point in its universe to another.

Flamm's Wormhole

In 1916, just one year after Einstein formulated his general relativistic laws of physics, Ludwig Flamm in Vienna discovered a solution of Einstein's equations that describes a wormhole (though he did not call it